

Haritha Hospitals — Migration Guide

Project: haritha-hospitals **Target:** pberpqa.duckdns.org (post 2026-08-21 rollback) **Source data:** masters/ (19 CSV files — canonical, idempotent) **Git repo:** origin/main at a0d1be9 (includes rollback event) **Last updated:** 2026-08-21 11:40 IST **Author:** Agent C (work-flow+migration+audit subagent)

□ **Credentials notice:** All credentials in this guide are intentionally **placeholder** values. - Administrator / <from site_config> — your site's Administrator user (from site_config.json) - <from ADMIN_PASSWORD env> — ERPNext admin password from .env (ADMIN_PASSWORD) - <from .env MYSQL_ROOT_PASSWORD> — MariaDB root password from .env (MYSQL_ROOT_PASSWORD)

Never commit real credentials. This repo uses placeholders throughout.

1. Goal

Redeploy Haritha Hospitals on pberpqa.duckdns.org (current env post-2026-08-21 rollback). The pberpqa.duckdns.org environment (9 containers) was destroyed at 10:11-10:18 IST. Company "Haritha Hospitals" was recreated on pberpqa at 10:59 IST. This guide covers the full redo of Phases 2-5 on the new environment.

2. Prerequisites

- pberpqa.duckdns.org running, HRMS 16.5.0 installed (pinned per Lesson #44)
 - pberp_hospital custom app installed (already done)
 - Haritha Hospitals company created (already done 10:59 IST on pberpqa)
 - CSV masters in masters/ (19 files — canonical source, idempotent)
 - Git history at origin/main (includes rollback event at a0d1be9)
 - VPS access: vijay@144.217.163.228 (for backup cron deployment)
 - Offsite backup target: venkat@135.125.196.35
-

3. Steps

Step 1: Phase 2 Redo — Site Setup

Target: pberpqa.duckdns.org (already has HRMS 16.5.0 + pberp_hospital app + Company)

On VPS (vijay@144.217.163.228) - if using pberpqa compose project
docker compose -p pberpqa ps *# Verify 9 containers running*

Verify site exists and apps installed
bench --site pberpqa.duckdns.org list-apps
Should show: frappe, erpnext, hrms (v16.5.0), payments, pberp_hospital

Verify Company exists
bench --site pberpqa.duckdns.org console
>>> frappe.get_doc("Company", "Haritha Hospitals")

If site needs recreation:

```
bench new-site pberpqa.duckdns.org \  
  --admin-password "<from ADMIN_PASSWORD env>" \  
  --mariadb-root-password "<from MYSQL_ROOT_PASSWORD env>" \  
  --no-mariadb-socket
```

```
bench --site pberpqa.duckdns.org install-app erpnext  
bench --site pberpqa.duckdns.org install-app hrms # v16.5.0 pinned  
bench --site pberpqa.duckdns.org install-app payments  
bench --site pberpqa.duckdns.org install-app pberp_hospital # custom app
```

Custom fields fixtures (already in pberp_hospital app):

```
bench --site pberpqa.duckdns.org migrate  
# Fixtures auto-applied on migrate
```

Holiday List (if not already loaded):

```
bench --site pberpqa.duckdns.org execute haritha.utils.load_holidays
```

Step 2: Phase 3 Redo — Data Import

Import all 19 CSV masters from masters/ in dependency order. The CSVs are idempotent — re-running is safe.

1. Foundation entities (L1)

```
bench --site pberpqa.duckdns.org execute haritha.import.departments  
bench --site pberpqa.duckdns.org execute haritha.import.employment_types  
bench --site pberpqa.duckdns.org execute haritha.import.leave_types  
bench --site pberpqa.duckdns.org execute haritha.import.designations
```

2. Shift Types (L2)

```
bench --site pberpqa.duckdns.org execute haritha.import.shift_types
```

3. Employees (210 records)

```
bench --site pberpqa.duckdns.org execute haritha.import.employees
```

4. ERPNext defaults cleanup (H1.5 + H1.5b)

```
bench --site pberpqa.duckdns.org execute haritha.cleanup.defaults
```

5. Shift Assignments (5,317 records) - background jobs

```
bench --site pberpqa.duckdns.org execute haritha.import.shift_assignments
```

6. Attendance (6,300 records) - raw SQL bulk insert

```
bench --site pberpqa.duckdns.org execute haritha.import.attendance
```

7. Employee Checkins (12,562 records) - 25 batches

```
bench --site pberpqa.duckdns.org execute haritha.import.employee_checkins
```

8. DB cleanup - force delete 5 X-HH Department variants

```
bench --site pberpqa.duckdns.org execute haritha.cleanup.xhh_departments
```

Validation (run after each step):

```
bench --site pberpqa.duckdns.org execute haritha.validate.counts  
# Should match: Dept=36, Desig=48, EmpType=6, LeaveType=7, ShiftType=25, Emp=210, ShiftAss
```

Step 3: Phase 4 Redo — Workflow Testing

```
# J-1: nginx config already in place on pberpqa
# Verify websocket proxy
curl -I https://pberpqa.duckdns.org/socket.io/
# Should return HTTP 200/101 with Upgrade header
```

```
# J-2: Worker fix verification
```

```
docker compose -p pberpqa exec queue-short python -c "import hrms; import payments; print(
```

```
# K-1: Smoke + functional tests
```

```
bench --site pberpqa.duckdns.org run-tests --module haritha.tests.test_smoke
```

```
# K-2: CRUD + payroll integration
```

```
bench --site pberpqa.duckdns.org run-tests --module haritha.tests.test_crud_payroll
```

```
# K-3: HRMS integration
```

```
bench --site pberpqa.duckdns.org run-tests --module haritha.tests.test_hrms_integration
```

Expected results (from rollback verification): - K-1: PASS (auth, CRUD, entity counts) - K-2: PASS (Shift Assignment CRUD, payroll integration) - K-3: PASS (shift/leave/holiday/process_attendance)

UI Smoke Test (manual required): 1. Open <https://pberpqa.duckdns.org> in real browser (Chrome/Firefox) 2. Login as Administrator 3. Verify Desk loads (not stuck on splash) 4. Navigate to Shift Assignment list — verify 5,317 records 5. Create test Shift Assignment via UI 6. Verify Employee Checkin list loads

Step 4: Phase 5 Redo — Production Readiness

```
# On VPS (vijay@144.217.163.228)
```

```
# Deploy backup script
```

```
mkdir -p /home/vijay/scripts /home/vijay/logs /home/vijay/backups
```

```
cat > /home/vijay/scripts/pberp_backup.sh << 'EOF'
```

```
#!/bin/bash
```

```
set -euo pipefail
```

```
TIMEOUT=900
```

```
BACKUP_DIR="/home/vijay/backups"
```

```
OFFSITE_HOST="venkat@135.125.196.35"
```

```
OFFSITE_DIR="/home/venkat/pberp_backups"
```

```
SITE="pberpqa.duckdns.org"
```

```
mkdir -p "$BACKUP_DIR"
```

```
TIMESTAMP=$(date +%Y%m%d_%H%M%S)
```

```
BACKUP_FILE="pberpqa_backup_${TIMESTAMP}.tar.gz"
```

```
timeout $TIMEOUT bench --site "$SITE" backup --with-files \
```

```
--backup-path "$BACKUP_DIR/$BACKUP_FILE"
```

```
BACKUP_EXIT=${PIPESTATUS[0]}
```

```
if [ $BACKUP_EXIT -eq 0 ]; then
```

```
    scp "$BACKUP_DIR/$BACKUP_FILE" "$OFFSITE_HOST:$OFFSITE_DIR/"
```

```
    find "$BACKUP_DIR" -name "pberpqa_backup_*.tar.gz" -mtime +14 -delete
```

```
    ssh "$OFFSITE_HOST" "find $OFFSITE_DIR -name 'pberpqa_backup_*.tar.gz' -mtime +30 -del
```

```
    echo " Backup complete: $BACKUP_FILE"
```

```

else
    echo " Backup failed with exit code $BACKUP_EXIT"
    exit 1
fi
EOF

chmod +x /home/vijay/scripts/pberp_backup.sh

# Test run
/home/vijay/scripts/pberp_backup.sh

# Install cron (every 6 hours)
(crontab -l 2>/dev/null; echo "0 */6 * * * /home/vijay/scripts/pberp_backup.sh >> /home/vi

# Verify cron
crontab -l

```

Pre-flight checklist: - [] Backup cron test run successful + offsite copy verified on venkat VPS - [] SSL certificates valid (expiry > 30 days) — check certbot certificates - [] All 9 containers healthy (docker compose -p pberpqa ps) - [] Database restore test on staging env (if available) - [] Monitoring: uptime, disk, memory alerts configured - [] Rate limiting on nginx (login, API endpoints) - [] Admin password rotated from default - [] setup_complete=1 in System Settings - [] desktop:home_page cleared for Administrator (or set to Desk)

Post-flight checklist: - [] First 24h: monitor error logs, slow queries, queue backlogs - [] Verify offsite backup arrives on venkat VPS - [] Confirm Auto Attendance cron processes correctly - [] Validate shift/attendance reports with real data - [] Document any new lessons learned in TRACKER.md

4. Rollback

If migration fails and you need to return to pre-migration state:

Option A: Restore from backup (preferred)

```

# On VPS
cd /home/vijay/backups
# Find latest backup
LATEST=$(ls -t pberpqa_backup_*.tar.gz | head -1)
bench --site pberpqa.duckdns.org restore "$LATEST" --force

# Offsite backup
scp venkat@135.125.196.35:/home/venkat/pberp_backups/pberpqa_backup_*.tar.gz /home/vijay/b
bench --site pberpqa.duckdns.org restore /home/vijay/backups/pberpqa_backup_*.tar.gz --for

```

Option B: Destroy and recreate site (nuclear)

```

bench drop-site pberpqa.duckdns.org --force
# Then re-run Step 1 (Phase 2) from scratch

```

Option C: Git revert (code only)

```
git revert a0d1be9 # Reverts rollback event commit  
# Note: This only reverts code, not database state
```

5. Lessons Applied

References to LEARNINGS.md and TRACKER.md Known Issues section:

Lesson #	Issue	Applied In Migration
44	HRMS v16.5.1+ broken by re-post_allowed_types	Pin HRMS to v16.5.0 in Step 1
47	Asset sync requires docker cp with / . syntax	nginx asset routing in Phase 4
79	bench backup --with-files needs timeout + \${PIPESTA-TUS[0]}	Backup script in Step 4
80	installed_apps in TWO places	Verified in Step 1 migrate
NEW	WebSocket Redis Socket-ClosedUnexpectedlyError	Health check + auto-restart in compose
NEW	Frappe ws needs Upgrade: websocket header	nginx config in Phase 4 J-1
NEW	Force-delete via direct DB DELETE	X-HH Department cleanup in Step 2
NEW	Headless browser unreliable	Manual browser test in Step 3

6. Quick Reference

```
# Full redo (single command sequence)
```

```
# Step 1: Site setup
```

```
bench --site pberpqa.duckdns.org migrate
```

```
# Step 2: Data import
```

```
bench --site pberpqa.duckdns.org execute haritha.import.all
```

```
# Step 3: Tests
```

```
bench --site pberpqa.duckdns.org run-tests --module haritha.tests
```

Step 4: Backup

```
/home/vijay/scripts/pberp_backup.sh
```

Verify counts

```
bench --site pberpqa.duckdns.org execute haritha.validate.counts
```

7. Support Files in Repo

File	Purpose
masters/*.csv	19 canonical CSV sources
fixtures/custom_field.json	14 custom fields (Rule #9)
scripts/import_*.py	Import scripts per entity
scripts/cleanup_*.py	Defaults + X-HH cleanup
scripts/validate_counts.py	Validation script
tests/test_smoke.py	K-1 tests
tests/test_crud_payroll.py	K-2 tests
tests/test_hrms_integration.py	K-3 tests
TRACKER.md	Phase log + Known Issues
DECISIONS.md	Architecture decisions
LEARNINGS.md	Consolidated lessons
WORKFLOW.md	This workflow (Phase 2-5 runbooks)

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